

An evidence-based multimodal fusion approach for predicting review helpfulness with human-AI complementarity

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Abstract: Online product reviews have been increasingly adopted by consumers to express their evaluation for products on e-commercial platforms. Consumers are encouraged to evaluate the review quality by helpful votes according to their human intelligence, and considerable machine learning approaches have been developed to help detect helpful reviews automatically. However, previous studies largely depend on machine intelligence and neglect the complementary strength between machine algorithms and human intelligence. This study proposes an evidence-based multimodal fusion approach for predicting the helpfulness of reviews by aggregating the human intelligence from user votes and machine intelligence from texts and images. Review helpfulness is measured by a belief distribution and represented by the evidence obtained from each modality data. An evidential reasoning rule-based fusion strategy is proposed to combine the hybrid intelligence considering their complementarity. A new dataset is also developed for the task of multimodal review helpfulness prediction by crawling online product reviews from Amazon.com and annotated by human scoring. Experimental results on the dataset show that the proposed approach for predicting review helpfulness has superior predictive power and outperforms existing methods. This study also contributes to e-commerce by providing an enhanced method for predicting review helpfulness from both human and machine intelligence.

Keywords: review helpfulness; multimodal fusion; evidential reasoning; hybrid intelligence

Highlights

1. An evidence-based multimodal fusion approach for predicting the helpfulness of reviews is proposed.
2. The complementary strength between human and machine algorithms can improve prediction performance.
3. Human intelligent from user votes is firstly formulated as a piece of evidence with a certain reliability for predicting review helpfulness
4. A new dataset is developed for multimodal review helpfulness prediction tasks.
5. Minimizing the judgment biases of machine and human intelligence from a local perspective can achieve superior prediction power.

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1.Introduction

The online product reviews are important sources of information in facilitating consumers' purchase decisions. As a form of word-of-mouth, online review can provide sufficient purchasing information that may influence consumers' behavior in e-commercial contexts (Xiao et al., 2022). However, the number of reviews for one product may easily go beyond the customers' capability of reading and processing all the reviews with different levels of quality (Guo et al., 2021). Hence, voting mechanism is usually adopted by e-commercial platforms, such as Amazon.com and Jindong.com, to encourage consumers to evaluate the helpfulness of reviews by voting. Consumers understand the contents and contexts related to reviews comprehensively and evaluate review helpfulness by human intelligence. The reviews with high helpful votes will be ranked on the top and can be found easily by consumers before they make purchase decisions.

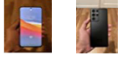
However, the helpful votes from human intelligence is likely to be biased and the review helpfulness cannot be effectively measured in current voting mechanism. There are many greatly-helpful reviews without any votes in e-commercial platforms. The reason may be that a lot of consumers do not vote even if they have read the review and regarded it as helpful. Statistics show that 62.2% reviews do not receive any vote and 91.7% reviews received no more than 10 votes (Guo et al., 2021). Moreover, most of helpful votes are concentrated on the small proportion of rich reviews that had received at least 10 votes (Liu et al., 2008). Hence, a great portion of reviews fail to receive sufficient votes. For example, there are three reviews related to iPhone products downloaded from Amazon.com, as shown in **Table 1**. The first review received 4 helpful votes. Owing to the less information in its content for making purchase decisions, it is regarded as unhelpful by human scoring in user studies. On the contrary, the second review with zero votes and the third one with two votes are considered as helpful since many aspects related to products are mentioned in their contents.

Considerable econometric models and machine learning approaches have been proposed to automatically assess the helpfulness degrees of reviews in response to the voting bias. They can be categorized into two groups. One stream of research analyzes potential determinants that affect review helpfulness from the explanatory perspective. Many theoretically motivated factors that may influence the helpfulness of online reviews have been explored from extensive empirical findings in previous studies. Econometric models are developed to investigate the causal relationships between the determinants and review helpfulness (Fresneda & Gefen, 2019). Since the explanatory factors and corresponding models are far from sufficient to achieve desirable prediction performance, the other research stream is centered on machine learning techniques for helpfulness prediction, such as support vector machines (Guo et al., 2021) and deep learning networks (Olmedilla et al., 2022). The deep semantic features extracted from review texts are used for predicting helpfulness by the convolutional neural network (CNN) or the long-short-term memory (LSTM) network (Mitra & Jenamani, 2021). The ground truth of the helpfulness adopted for training a machine learning model traditionally comes from helpful votes by specifying a threshold for the percentage of positive votes (Guo et al., 2021) or for the number of votes (Olmedilla et al., 2022).

Although machine intelligence above has achieved high efficiency when predicting review helpfulness, it largely depends on the training data and prediction performance is still to be improved. There are a lot of complicated multimodality information including review texts and corresponding images. Although human judgment from votes are likely to be biased, it has been an important information cue for evaluating the quality of reviews for consumers. Moreover, machine and human intelligence have different capabilities of processing information and evaluating helpful reviews. Prior studies have demonstrated that humans and machine algorithms can have complementary strengths and lead to better

performance than combinations of human or machine predictions alone (Steyvers et al., 2022). However, both machine and human intelligence have limitations and reliabilities at a certain extent. How to symbiotically combine machine intelligence from multimodal data and human judgment from user votes considering their individual reliabilities need to be further investigated.

Table 1 Examples of the reviews with different presentation formats

Reviews	User votes	Human Scoring(0-10)
★★★★★ upgrade from s10e do not understand why people complain about 21u, it's a great phone  4 people find this helpful	4	0.9
★★★★★ awesome camera incredible camera with telephoto. fast processing. only downside is <u>samsung</u> bloat.  0 people find this helpful	0	2.4
★★★☆☆ battery doesn't even go last 1 day. i had 2019 version and even after 2 years of use, the battery lasted 1.5 days. This one barely go 1 day.  2 people find this helpful	2	4.5

This study aims to present an aggregation approach for enhancing the prediction performance of review helpfulness by combining the machine judgment from review texts and images, and human judgment from user votes. The user votes, texts and images provided by consumers can be regarded as three modalities where a modality refers to the way in which something happens or experienced (Baltruaitis et al., 2019). The modality data of helpful votes represents the evaluation results from human intelligence. The modalities of both texts and images reflect the essence of reviews and can be used for evaluating the helpfulness of reviews by machine algorithms. The Dempster-Shafer (DS) evidence theory and evidence combination rules are often used for aggregating machine classifiers (Liu et al., 2018). Therefore, this study presents a multimodal late fusion framework based on the DS theory, and jointly combine the human and machine intelligence for predicting review helpfulness. A new multimodal dataset is also developed for the task of review helpfulness prediction.

The contributions of this research are summarized as follows. (1) An evidence-based multimodal fusion (EMF) approach is proposed for predicting review helpfulness by aggregating human judgment and machine judgment from different modalities. It is the first time that the helpfulness degrees of reviews are predicted by hybrid collective intelligence. (2) An evidential reasoning rule-based fusion strategy is proposed for combining the human and machine intelligence considering their complementarity. (3) A new dataset is developed for the task of multimodal review helpfulness prediction by crawling data from Amazon.com. The ground truth of helpful degrees is obtained from human scoring. Experimental results show that the proposed approach has superior predictive power compared with the state-of-the-art end-to-end neural networks and other aggregation methods.

The remainder of this paper is organized as follows. Related work is reviewed in Section 2. Then, an evidence-based multimodal fusion approach is proposed in Section 3 for predicting review helpfulness. The experiments are designed in Section 4, and Section 5 provides experimental results and discussion. The paper is concluded in Section 6.

2.Related work

2.1 Determinants of review helpfulness

The features or determinants that may affect the helpfulness of reviews are usually identified from an explanatory perspective. They can be mainly categorized as review-related factors and reviewer-related factors. The review-related factors include stylistic, semantic, readability and sentiment features. Abbasi and Chen (2008) utilized a kernel theory of systemic functional linguistics and identified linguistic features from four categories including structure, lexicon, syntactic and semantic groups. Cao et al. (2011) used text mining techniques to examine stylistic characteristics, and demonstrated that semantic features are important when predicting the number of helpfulness votes. Ghose and Ipeirotis (2011) extended the research on readability and computed a set of metrics for measuring how easy is for a consumer to read a review. Yin et al. (2014) explored the effects of negative emotions including anxiety and anger on the readers' perceived helpfulness and demonstrated the importance of examining discrete emotions in online reviews. Reviewer-related factors are often referred as reviewers' social features. Chen and Tseng (2011) suggest that reviewers' reputation has great impact on review quality. Zheng (2013) selected six features to capture the social characteristics of reviewers, including reviewers' rank in the last few days, top flag, total helpfulness votes received, total review number, historical reviewers' rank and the number of personal information items. Besides, extra features related to review contents are introduced with the support of text-mining techniques, such as information gain, lexical similarity between reviews and product descriptions of other reviews, and information entropy increments (Fresneda and Gefen 2019).

2.2 Machine learning approaches for predicting review helpfulness

Many machine learning models and deep neural networks have been developed for predicting the helpfulness degrees of reviews. They are often trained on a set of reviews with sufficient votes and aim to formulate the relationships between review features and helpfulness degrees. The helpfulness prediction scenario is often formulated as a classification or regression problem.

In classification context, review helpfulness is categorized into two or multiple classes according to received votes. For example, Chen and Tseng (2011) classified product reviews into five classes according to their quality level including high, medium, low, duplicate and spam. Liu (2007) defined four categories of review quality representing different values of the reviews to users' purchase decision including best, good, fair, and bad reviews. In a binary classification, a threshold is often specified for the percentage of positive votes or thumb-up votes in identifying whether a review is helpful or not. The review with the percentage of positive votes or thumb-up votes above the specified threshold is treated as helpful one. Otherwise, it is regarded as unhelpful. Ghose and Ipeirotis (2011) specified the threshold as 60% for the percentage of positive votes. Ma et al. (2018) labeled the reviews with at least one helpful count as helpful ones, and proposed a multi-modality feature fusion model to demonstrate the usefulness of deep learning techniques in processing and analyzing large quantities of image data. Du et al. (2021) labeled a review as helpful if it received at least two votes, and proposed an end-to-end neural architecture to capture the interaction between reviews and their neighbors. Olmedilla et al. (2022) developed a one-dimensional CNN for classifying reviews into helpful, neutral and not helpful categories, and modelling the online review helpfulness with contextual characteristics. Xiao et al. (2022) regarded the review with more than one votes as helpful, and exploited the interaction between images and texts when predicting review helpfulness.

Since the classification approaches cannot directly rank reviews, a large amount of machine learning approaches focus on predicting the helpfulness degree of each individual review. In the regression context, the helpfulness of reviews can be measured based on the percentage-based measurement or probability-based measurement. The percentage-based measurement is often used to simply evaluate the helpfulness degree of a review with the percentage of positive votes, such as X/Y . Generally, X

represents the helpful votes that a review received and Y may be the total votes that the review received (Liu et al., 2008). Mitra and Jenamani (2021) predicted the helpfulness degrees by the LSTM and a dual-CNN for extracting semantic features and combined them with the syntactic features of reviews. Since the widely adopted X/Y measurement could lead to highly biased results, Guo et al. (2021) proposed a probability-based helpfulness measurement and regarded consumers' voting behavior as a random event from a probability perspective. The helpfulness degree is defined as a probability that the consumer regards this review helpful, and predicted by support vector machines (Liu 2007, Chen and Tseng 2011).

Although previous machine learning approaches have trained many complicated models from a predictive perspective for review helpfulness, it is validated that these approaches have systematic weaknesses and blind spots. Especially, state-of-the-art deep learning classifiers for images and texts may fail in unpredictable and unexplainable ways. Since the user votes are from human judgment with human's comprehension for review texts and images, they should be considered and aggregated with machine algorithms for review helpfulness prediction.

2.3 Multimodal fusion approaches

Multimodal machine learning aims to build models that can process and reason information from multiple modalities. Integrating the information from multiple modalities that observe the same phenomenon can achieve robust predictions (Baltrusaitis, 2019). The texts, images, and helpful votes related to consumers' reviews can be regarded as three modalities of reviews, which provide useful information for understanding the helpfulness of reviews from three local perspectives. The complementary information that is not visible in individual modality may be captured from multiple modalities. Moreover, the multimodal prediction system can still operate even when one of modalities is missing (D'mello and Kory, 2015).

The decision-level fusion or late fusion approaches have been widely used in emotion recognition and multimedia analysis, among state-of-the-art multimodal fusion approaches. They are not directly dependent on specific machine learning methods and easy to implement using unimodal machine learning methods. The prediction scores available from each modality are merged by a fusing mechanism at a decision level. Many fusion mechanisms have been proposed to combine different classifiers, such as majority voting (Morvant et al., 2014), average (Shutova et al., 2016), and weighting based on a learned model (Moreno-seco, 2006). The traditional decision-level fusion mechanism is associating probability scores to each possible decision from each modality and then using them as input vectors for a probabilistic inference system. Recently, Dempster-Shafer evidence theory-based fusion approaches have been proposed to combine machine classifiers for dealing with the uncertain information in fusion processes. For example, Liu et al. (2018) proposed a weighted DS fusion rule to leverage the complementary knowledge among classifiers for improve the accuracy. Yang et al. (2019) proposed an evidential reasoning rule for aggregating the base classifiers' predictions from random subspaces and alleviating the biases in learning processes. However, the complementarity among classifiers is still to be further investigated, especially when machine and human classifier are aggregated for predicting review helpfulness together.

3. The evidence-based multimodal fusion approach

An evidence-based multimodal fusion approach is proposed in this study to predict review helpfulness using the reviews' different presentation formats. The helpfulness degrees are predicted not only based on consumers' voting behavior but also on the texts and images related to reviews. The predicted final helpfulness degree is the aggregation of four categories of evidence obtained from consumers' votes, theoretical features, textual and visual contents, respectively.

3.1 The proposed framework

According to the multimodality theory, the fusion of information from different sources may reduce the

overall uncertainty and increase the robustness of a prediction system. Decision-level fusion approaches have achieved great prediction results in multimedia analysis owing to its usage of different prediction models for each modality (Hermessi et al., 2021). Therefore, an evidence-based decision-level fusion approach is proposed in this study for predicting review helpfulness not only by human judgments, but also by machine intelligence from texts and images. The overall framework of the approach is shown in **Figure 1**.

First, structural features of reviews including information quality and review consistency are constituted for describing a review from a global perspective, according to the elaboration likelihood model (ELM) theory. Then the Bi-LSTM and CNN network are developed for learning the feature representation of texts and images, respectively. Three machine algorithms are proposed to predict review helpfulness from handcraft features, texts and images and generate three kinds of evidence, respectively. Meanwhile, human judgment is also formulated as a piece of evidence. All the evidence is aggregated by an evidential reasoning-rule based fusion strategy for predicting the helpfulness of reviews.

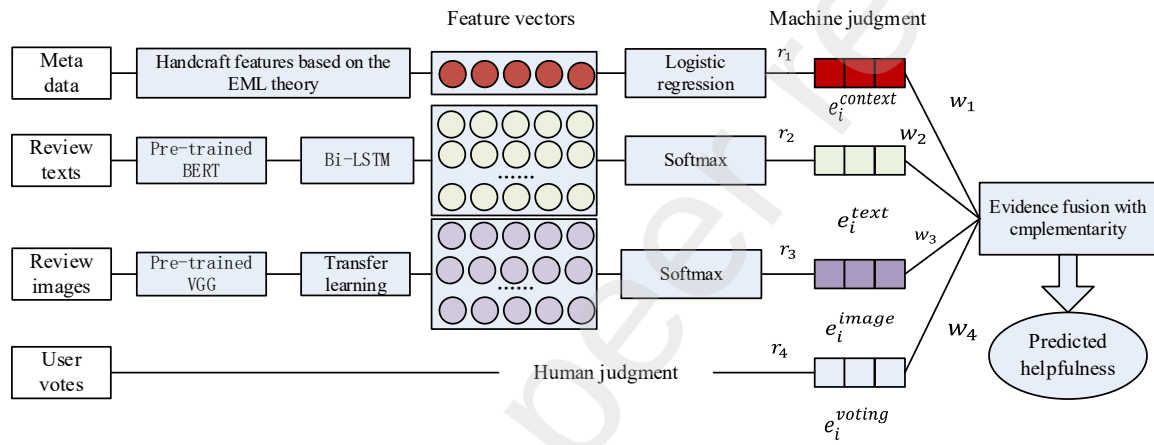


Figure 1. The framework of the proposed approach

3.2 Human intelligence from user votes

The number of helpful votes received by reviews reflects users' behavior. When using the helpful votes for predicting review helpfulness, the human judgment from votes should be formulated as a piece of evidence and be consistent with machine judgment from texts and images. Therefore, we have the definitions of review helpfulness and judgment bias in the Dempster-Shafer scheme.

Definition 1 (helpfulness degree): Supposed that a frame of discernment $\Omega = \{H_1, H_2\} = \{\text{helpful}, \text{unhelpful}\}$ is defined to represent two states of a review S_i ($i = 1, \dots, N$). The helpfulness degree can be defined as a belief distribution where a consumer would vote this review helpful at a probability level.

$$e_i = \{(\text{helpful}, P_i), (\text{unhelpful}, 1 - P_i)\}, i = 1, \dots, N$$

where N is the number of review samples, and P_i ($i = 1, \dots, N$) is the probability that a customer would regard the review S_i ($i = 1, \dots, N$) as helpful. When multiple pieces of evidence for measuring review helpfulness are obtained from different presentation formats of reviews, they can be combined to achieve a final helpfulness degree.

Definition 2 (ground truth of review helpfulness): The ground truth of review helpfulness can be labeled as helpful or not helpful according to the specified threshold for user votes. After the specified is set, the ground truth of review helpfulness can be denoted as a piece of evidence e_G .

$$e_G = \{(\text{helpful}, Q_i), (\text{unhelpful}, 1 - Q_i)\}, i = 1, \dots, N, \text{ and } Q_i = \begin{bmatrix} 1, & \text{helpful} \\ 0, & \text{unhelpful} \end{bmatrix}.$$

Definition 3 (judgment bias): Since there is uncertain information in both human and machine algorithms, the judgment bias JB_i may exist in both human and machine intelligence. It can be formulated as the dissimilarity between the predicted helpful degree e_i ($i = 1, \dots, N$) and the ground truth $e_{G,i}$ ($i = 1, \dots, N$) of the review S_i ($i = 1, \dots, N$).

$$JB_i = \|e_i - e_{G,i}\|_2, i = 1, \dots, N$$

Suppose that the review S_i ($i=1, \dots, N$) has received V_i ($i=1, \dots, N$) votes. A threshold $[\delta_1, \delta_2]$ can be set for judging whether the review is helpful or not. The helpfulness degree p_i^v ($i=1, \dots, N$) of the review S_i ($i=1, \dots, N$) predicted from human judgment can be defined as a piece of evidence, shown in Eq. (1).

$$e_i^{voting} = \{(helpful, p_i^v), (unhelpful, 1 - p_i^v)\}, i = 1, \dots, N.$$

$$p_i^v = \begin{cases} 1, & V_i > \delta_2, \text{ completely helpful} \\ \frac{V_i}{\delta_2}, & \delta_1 \leq V_i \leq \delta_2, \text{ helpful in some extent} \\ 0, & V_i < \delta_1, \text{ unhelpful} \end{cases}, i = 1, \dots, N \quad (1)$$

3.3 Machine intelligence from theoretical features

Theory-driven research in review helpfulness has dominated information systems research for decades. Among them, the ELM theory provides a theoretical framework for understanding the effectiveness of persuasion in computer-mediated and online environments (Sussman and Siegal, 2003). It has been used as theoretical foundation to examine the factors that affect the perceived credibility of online reviews (Cheung et al., 2012). The ELM theory states that two routes of information affect people's attitude toward persuasive information, including the central route mainly dealing with the contents and topics of the message itself, and the peripheral route dealing with the peripheral information related to the message sources and their characteristics. According to the domain knowledge in review helpfulness and the ELM theory, information quality is identified as a central cue and information consistency as a peripheral cue in this study for selecting features to predict the helpfulness of reviews.

3.3.1 Feature extraction based on the ELM theory

In the context of review helpfulness prediction, information quality and review consistency are usually regarded as the cues from the central route and the peripheral route, respectively. When a consumer processes a review through the central route, he may carefully consider the information quality presented in the review and exert more efforts to evaluate it with a high level of elaboration. In contrast, the peripheral route involves a low level of elaboration. Simple heuristic cues such as review consistency can be used to assess the credibility of a review (Cheung et al., 2012). The features and their corresponding definitions are detailed in Table 2.

Table 2 The theoretically-motivated features

Factors	Features	Definition	References
Information quality	Review depth	Review length, the number of nouns, verbs, personal pronouns, pronouns, adverbs, adjectives	Mudambi and Schuff (2010). Hong et al. (2017)
	Readability	FKRE, GFI, ARI	Korfiatis et al. (2012)
	Subjectivity	Number of positive words and negative words, sentiment scores	Ghose and Ipeirotis (2011), Hu et al. (2014)
	Relevancy	text-image similarity and rating-sentiment consistency	Liu et al.(2020), Shan et al.(2021)
	Content complexity	Text complexity and image complexity	Shin et al., 2020
Information consistency	Content consistency	Deviation of the occurrence counts of nouns, verbs, personal pronouns, pronouns, adverbs, adjectives; topic consistency	Shan et al. (2021), Shi et al., (2016), Shin et al. (2021)
	Sentiment deviation	The deviation of sentiment scores	Fresnada and Gefen (2019), Hu et al.(2014)

(1) Information quality features from a central route

Information quality has been identified as an important antecedent of review helpfulness in the literature. It refers to the consumers' subjective perception of the information in the persuasive message as strong and cogent. As a central cue, information quality has also been shown to be crucial for knowledge adoption in organizations (Sussman & Siegal, 2003). In the context of online reviews, a review with a high information quality should be perceived as helpful. Four information quality features are identified from the central route, including review depth, readability, subjectivity, relevancy, and content complexity.

Review depth measures the number of words in a review (Mudambi and Schuff, 2010). We quantify review depth using the number of nouns, verbs, personal pronouns, pronouns, adverbs, adjectives and review length. Readability is defined as the ease of understanding or comprehension due to the style of writing. The following three formulas are often used as readability features: Automated readability index (ARI), Gunning fog index (GFI) and Flesh-Kincaid Reading Ease Score (FKRE). The readability is also called the accessibility that a review can be read easily by a large number of users. Subjectivity describes consumers' sentiments and feelings about products or services. The number of positive and negative words, the sentiment scores of reviews are quantified in this study to measure the subjectivity of a review.

Relevancy represents the inner consistency of different formats of reviews. It has been employed to predict the quality and helpfulness of knowledge solutions in online communities (Liu et al., 2020). Relevancy involves not only the similarity between the text and corresponding images, but also the consistency between ratings and emotions expressed in a review text, which are instrumental in helping consumers perceive the quality of reviews (Shan et al., 2021). To measure the similarity between images and texts, the pre-trained CNN model is often used to obtain the representation of images p_{image} and a word embedding technique, word2vec is for the vector of texts p_{text} . The rating-sentiment consistency can be defined as the absolute difference between the rating of a review and content sentiment. The sentiment and rating are both normalized as z-scores.

$$text_image_similarity = \frac{p_{image} \cdot p_{text}}{\|p_{image}\| \cdot \|p_{text}\|}$$

$$rating_sentiment_consistency = |z_score_{rating} - z_score_{sentiment}|$$

Content complexity refers to the structural variations in both text and image contents. It captures the high-level semantics embedded in an image or a text. A topic-level text complexity measure can be used to capture the diversity of topics covered in the text. The latent Dirichlet approach (LDA) topic modeling approach is employed to capture the topics of the overall text at a macro level (Gong et al., 2018). Texts covering multiple topics are considered semantically complex. The topic-level text complexity is defined as the Shannon diversity index, where $d1$ is the number of topics and p_i is the probability of the i th topic in a given review.

$$text_complexity = -\sum_{i=1}^{d1} p_i \log(p_i), \sum_{i=1}^{d1} p_i = 1$$

Meanwhile, object-level complexity is proposed to assess visual structure variations in terms of specific objects, shapes, and arrangements (Shin et al., 2020). The images from social media posts are input to the pre-trained CNN model to obtain their confidence scores in 1,000 object categories. The Shannon diversity index is also employed to measure the variety in the confidence scores for an image, where $d2$ is the 1000 and q_i is the confidence score of the i th category in a given image.

$$image_complexity = -\sum_{i=1}^{d2} q_i \log(q_i), \sum_{i=1}^{d2} q_i = 1$$

(2) Information consistency features in a peripheral route

Information consistency refers to the extent to which information in a review is consistent with information in other reviews. It has a positive effect on review credibility (Cheung et al., 2012). Users are more likely to accept a viewpoint that is consistent across most reviews, and more likely to be skeptical toward a viewpoint that is voiced by a comparatively small number of reviewers. Three consistency factors including content consistency, topic consistency and sentiment deviation, are identified as peripheral cues in evaluating review helpfulness.

Content consistency evaluates whether a review content is deviated from the average content of all reviews with the same star ratings for same products (Shan et al., 2021). Six content features are selected for measuring the content consistency among reviews from the numbers of noun, verb, personal pronoun, pronoun, adverb, adjective words. For example, when the noun count of the review j ($j=1, \dots, N$) with a rating i ($i=1, \dots, 5$) is denoted as f_i^j , the content consistency feature in the perspective of nouns is represented as $f_{noun} = \frac{|f_i^j - \bar{f}_i|}{\sigma_i}$, where $\bar{f}_i$ and σ_i is the mean and standard deviation of f_i^j across all reviews with the same star rating i ($i=1, \dots, 5$), respectively.

Meanwhile, the topic-based text similarity measure is well-adopted in the IS research to evaluate content consistency between texts (Shi et al., 2016, Shin et al., 2021). It evaluates whether a review is similar to or distinct from the usual or average content of all reviews related to a product. The vector of a review text c_i is set as the corresponding topic distribution computed via the LDA. The average content of the review is defined as $c_i^{average}$ and Ω_i is the set of reviews created by the customers prior to the review c_i .

$$text_{similarity} = \frac{c_i^{average} \cdot c_i}{\|c_i^{average}\| \cdot \|c_i\|}, \text{ where } c_i^{average} = \frac{1}{|\Omega_i|} \sum_{j \in \Omega_i} c_j$$

Sentiment deviation measures the degree that the sentiment of a review is not neutral (Fresnada and Gefen 2019). In predicting review helpfulness, a two-sided review may be perceived as more unbiased than a one-sided review. A one-sided review contains either positive or negative product comments, whereas a two-sided review contains both positive and negative comments on a product. Consumers usually regard the reviews with moderately negative and moderately positive sentiments to be helpful (Hu, et al., 2014). It can be measured by the absolute value of the sentiment score of the review minus the value of the median score of all the reviews.

3.3.2 Evidence acquisition

Logistic regression is used in this studies for predicting the helpfulness degrees according to the handcraft features above. The prediction result is formulated as a piece of evidence. The logit function is used to represent the probability values of helpfulness from 0 to 1. Suppose that $Y \in \Omega = \{H_1, H_2\}$ is a class variable, and $X = (x_1, \dots, x_M)$ are feature vectors, such as information quality and consistency features. Let $p(x)$ denotes the probability that $Y=H_1$ given that $X=x$. The binary logistic regression model is shown as following:

$$\ln \left(\frac{p(x)}{1-p(x)} \right) = \beta^T x + \beta_0, \quad p(x) = \frac{1}{1 + \exp[-(\beta^T x + \beta_0)]}$$

where $\beta \in R^M$ and $\beta_0 \in R$ are two parameters representing a vector of weights and an intercept, respectively. Given a learning set $\{(X_1, Y_1), (X_2, Y_2), \dots, (X_N, Y_N)\}$, the two parameters are estimated by maximizing the conditional log-likelihood:

$$\ell(\beta, \beta_0) = \sum_{i=1}^N y_i \ln p(X_i) + (1 - y_i) \ln (1 - p(X_i))$$

where $y_i=1$ if $Y_i = H_1$, and otherwise, $y_i=0$. The logistic function presents a way to take a sample x and compute the probability $p(y=1|x)$. If a review S_i ($i=1, \dots, N$) is predicted to be helpful at the probability of p_i^k from the logistic regression model, then the predicted review helpfulness of the review from the handcraft features can be formulated as a piece of evidence:

$$e^{H-feature} = \{(helpful, p_i^k), (unhelpful, 1 - p_i^k)\}, i = 1, \dots, N. \quad (2)$$

3.4 Machine intelligence from texts and images

Deep learning models have been increasingly adopted for automatically extracting features from textual reviews. The machine learning-enabled features shown substantial improvement in predicting persuasiveness of social media content over the handcrafted theory-based features (Shin et al., 2020). Therefore, a textual Bi-LSTM network and a visual CNN are developed in this study for predicting

review helpfulness from text and image contents, respectively.

3.4.1 Evidence acquisition from the textual Bi-LSTM network

A textual deep learning model is proposed in this study for predicting review helpfulness from texts. The structure of the model is comprised of three modules including a word embedding module for transforming a text into a vector, a Bi-LSTM network for feature extraction, and a classification module for evidence acquisition.

The word embedding module is developed for translating review texts into numerical representations. The pre-trained BERT (Bidirectional encoder representation from transformers) language model (Devlin, 2019) is adopted in this study to represent review texts by considering the context of a word from both the left and right sides of the word simultaneously. The review text S_i ($i=1, \dots, N$) can be transformed into a real-valued matrix $R_i \in \mathbb{R}^{d \times L}$ in the embedding layer, where d is the dimension of embedding vector and L is the maximum number of tokens that represent the words in a review text.

The Bi-LSTM network is adopted to learn the textual features in binary directions simultaneously including forwards and backwards. It can present each training sequential texts to two separate LSTM networks, both of which are connected to the same output layer (Yu et al., 2020). This means that the Bi-LSTM can provide each point in a given sequence with complete and sequential information about all words before and after the point. Both the forward path and backward path in each LSTM network are computed independently and their outputs are concatenated as an output vector of the Bi-LSTM network (Zhang, et al., 2018).

In the forget gate level, the LSTM memory cell decides what information will be forgot from the last cell state C_{t-1} by a sigmoid function $\sigma(\cdot)$. The output of last cell h_{t-1} and the input of current t time x_t are transformed by the sigmoid function into a number f_t between 0 and 1 (Hochreiter & Schmidhuber, 1997).

$$f_t = \sigma(W_f h_{t-1} + U_f x_t + b_f)$$

The zero represents completely throwing away it and the one represents keeping it completely. In the input gate level, both a sigmoid function $\sigma(\cdot)$ and a hyperbolic tangent function $\tanh(\cdot)$ are used to decide what new information will be stored in the cell state. The sigmoid function decides the values to be updated and transforms both the output of last cell h_{t-1} and the current input x_t into a vector i_t . The hyperbolic tangent function creates a vector of new candidate values $\tilde{C}_t$ between -1 and 1 that would be added to the cell state. Then the old cell state C_{t-1} is updated into a new cell state C_t by the equation:

$$i_t = \sigma(W_i h_{t-1} + U_i x_t + b_i), \quad \tilde{C}_t = \tanh(W_c h_{t-1} + U_c x_t + b_c), \quad C_t = f_t * C_{t-1} + i_t * \tilde{C}_t.$$

In the output gate level, the output information of the new cell state C_t is decided by both a sigmoid function and a hyperbolic tangent function. The sigmoid function decides what parts of the new cell state to be output and the tanh function pushes the values of C_t to be between -1 and 1. Then the output of the cell h_t is calculated by the equation:

$$o_t = \sigma(W_o h_{t-1} + U_o x_t + b_o), \quad h_t = o_t * g(C_t).$$

When a review text with L words is fed into the Bi-LSTM network, a text feature vector $v_{text} = (h_1, \dots, h_t)$ is obtained for representing the review. Then the text vector is transformed into a low-dimensional vector using a linear layer. A Softmax function is adopted as a classifier for predicting the helpfulness degree p_i^t from texts, shown in Eq. (3).

$$p_i^t = P(y = 1 | v_{text} W, b) = \text{softmax}(v_{text}^T W + b) \quad (3)$$

Therefore, the helpfulness degree predicted by the text related to the review S_i ($i=1, \dots, N$) is transformed into a piece of evidence e_i^{text} by the developed textual deep learning model.

$$e_i^{text} = \{(\text{helpful}, p_i^t), (\text{unhelpful}, 1 - p_i^t)\}, i = 1, \dots, N.$$

3.4.2 Evidence acquisition from the visual CNN

The visual content features related to online reviews are investigated through the CNN in this study. The CNN model can automatically discover the intricate structure of images and generate a robust representation of the image. It has served as a testbed for large-scale image classification systems (Krizhevsky et al., 2012). A pre-trained CNN is adopted first and then fine-tuned by transfer learning for image representation of reviews.

The VGG (Visual geometry group) model is adopted for predicting review helpfulness owing to its depth-increased structure with very small convolution filters. There are many pre-trained CNN models including VGG, ResNet, and others. A pre-trained model has the pre-defined architecture and randomly-initialized parameters which need to be trained again on a specific task. The representation output from the VGG model can generalize well to other datasets (Simonyan and Zisserman, 2015). It is pre-trained on the ImageNet dataset. By maximizing the multinomial logistic regression objective, its parameters are optimized to perform well on the ImageNet 1000-class natural image classification task. The pre-trained VGG model consists of multiple layers of neural networks including an input layer, convolutional layers and an output layer. Each convolutional layer transforms the representation of an image from the previous layer into a more abstract representation by a convolution operator. The output layer in the pre-trained VGG produces a distribution over the 1000 class labels.

Transferring the parameters of a pre-trained VGG and fine-tuning them to another model for a different task is a powerful and common practice in deep learning, namely transfer learning. It can take advantages of pre-trained models that have been trained on large datasets and achieve outstanding performance in small datasets (Oquab et al., 2014). The pre-trained CNN containing multiple convolutional layers and three fully-connected layers. The representation vector of an image obtained from the fine-tuned VGG model is kept and the final full connected layer is replaced by a new layer with the right number of outputs for predicting review helpfulness.

A Softmax function is also adopted as a classifier for predicting the helpfulness degree p_i^m from images, shown in Eq. (3).

$$p_i^m = P(y = 1 | v_{image}, W, b) = \text{softmax}(v_{image}^T W + b) \quad (4)$$

Therefore, the helpfulness degree predicted by the images related to the review S_i ($i = 1, \dots, N$) is transformed into a piece of evidence e_i^{image} by the developed visual deep learning model.

$$e_i^{image} = \{(\text{helpful}, p_i^m), (\text{unhelpful}, 1 - p_i^m)\}, i = 1, \dots, N$$

3.5 An evidential reasoning rule-based fusion strategy

Predicting review helpfulness of the review S_i ($i = 1, \dots, N$) from hybrid machine and human intelligence can be regarded as a combination of the four kinds of evidence obtained from the above models. The evidential reasoning (ER) rule, proposed by Yang and Xu (2013), is introduced in this study for aggregating the hybrid intelligence with consideration of their complementary strengths. The complementarity is represented by both the reliability and weight of evidence. The weight of evidence reflects the relative importance of machine and human intelligence, and the reliability of it measures the inherent predictive performance from each intelligence considering the uncertainty and bias in prediction processes.

3.5.1 Reliability of machine and human intelligence

Since the evidence for predicting review helpfulness comes not only from machine algorithms but also from consumers' voting behavior, a contextual reliability of evidence is defined in this study to represent the inherent uncertainty of the evidence. The contextual reliability is dependent on the uncertainties

existing in both machine learning models and consumers' voting behaviors.

Since the uncertainty of the evidence from machine judgment mainly comes from their prediction performance. The performance metrics for classification are generally calculated according to a confusion matrix. Four factors in the matrix include true positive (TP), true negative (TN), false positive (FP), false negative (FN). A helpful review is usually regarded as positive. Meanwhile, the uncertainty of the evidence from human judgment arises from voting bias. A threshold $[\delta_1, \delta_2]$ is set for judging whether the review with initial votes is helpful or not. When the threshold is set similarly to that in user study, a confusion matrix can also be constituted where the prediction results are from consumers' votes and the ground-truth labels are from user study.

Suppose that the reliability of evidence is formulated by r_j ($j = 1, \dots, J$) for representing the uncertainties measured by the confusion matrix. The contextual reliability of the evidence r_j ($j = 1, \dots, J$) is calculated by the prediction accuracy both from the machine classifiers and human classifier.

$$r_j = accuracy_j = \frac{TP_j \times TN_j}{TP_j + FP_j + TN_j + FN_j}, j = 1, \dots, J \quad (5)$$

3.5.2 Evidence fusion

Multiple pieces of evidence from the hybrid intelligence are combined by the evidential reasoning rule with the contextual reliability. Suppose that there are J pieces of evidence related to a review S_i ($i = 1, \dots, N$). They come from different machine algorithms and humans, independent from each other. A piece of evidence e_j ($j = 1, \dots, J$) can be profiled as:

$$e_j = \{(\theta, p_{\theta,j}) | \theta \in \Omega, \sum_{\theta \in \Omega} p_{\theta,j} = 1\}, j = 1, \dots, J, \Omega = \{H_1, H_2\}$$

When the reliability and weight of evidence are taken into consideration, a belief distribution with the weight w_j ($j = 1, \dots, J$) and the reliability r_j ($j = 1, \dots, J$) can be defined as:

$$m_s = \{(\theta, \tilde{m}_{\theta,j}) | \theta \in \Omega; (P(\Omega), \tilde{m}_{P(\Omega),j})\}$$

$$\tilde{m}_{\theta,j} = \begin{cases} 0 & \theta = \phi \\ c_j m_{\theta,j} & \theta \in \Omega, \theta \neq \phi \\ c_j(1 - r_j) & \theta \in P(\Omega) \end{cases}, c_j = 1/(1 + w_j - r_j), m_{\theta,j} = w_j p_{\theta,j}$$

Where $\tilde{m}_{\theta,j}$ represents the support degree from the evidence e_j for the hypothesis θ . The final helpfulness degree P_i can be formulated as a piece of evidence e_i^S ($i = 1, \dots, N$) obtained by combining all the evidence related to the review S_i ($i = 1, \dots, N$).

$$e_i^S = e_i^{voting} \oplus e_i^{structure} \oplus e_i^{ext} \oplus e_i^{image} = \{(helpful, P_i), (unhelpful, (1 - P_i))\} \quad (6)$$

Both the binary cross entropy and the judgment bias are combined as the loss function for the proposed EMF approach to learn the optimal weight of evidence. The binary cross entropy is a widely used alternative for the squared error in machine learning. When the output of a model is a probability distribution, cross entropy can measure the distance between the predicted distribution of the model and the original distribution from a global perspective. The judgment bias defined in this study reflects the dissimilarity between the evidence from the prediction model and that from the ground truth for each review. Compared with the cross entropy, the judgment bias measures the predicted error from a local perspective. The loss function in the learning process is as follows:

$$l(Y, P, w) = -\frac{1}{N} \sum_{i=1}^N (Y_i \log(P_i) + (1 - Y_i) \log(1 - P_i)) + \frac{1}{N} \sum_{i=1}^N \|e_i - e_{G,i}\|_2^2 \quad (7)$$

Where N is the number of training samples, Y_i is the ground-truth label of the i th ($i = 1, 2$) class that encoded with one-hot and P_i is the predicted probability of the i th class. The goal of the loss function is to compare how well the predicted review helpfulness matches the ground-truth label of data. The lower the loss, the more accurate the prediction model.

The stochastic gradient descent (SGD) optimizer is adopted to train the fusion model, just like the above

deep learning neural networks. The optimization algorithm is widely used in machine learning applications to find model parameters corresponding to the best fit between predicted probabilities and actual outputs. It has advantages of efficiency and ease of implementation.

3.5.3 Algorithm structure

The main components in the proposed EMF approach include evidence acquisition from hybrid intelligence and evidence fusion for predicting the helpfulness degrees. The computational details are shown in Algorithm 1. First, training dataset is acquired through the review and its different formats including texts, images and votes. The ground-truth label of each review is obtained by human scoring. Meanwhile, the ELM theory-based features including information quality and consistency are constituted from the reviews. Then, four kinds of evidence are obtained from both machine and human intelligence. The contextual reliability of the evidence is calculated by a confusion matrix and its weight is learned from data. Last, the predicted review helpfulness values and the weight of evidence are output.

Algorithm 1 Evidence acquisition and fusion in the EMF approach

Input: training review set including S_i ($i=1,\dots,N$) and received votes V_i ($i=1,\dots,N$)
Output: predicted review helpfulness values P_i ($i=1,\dots,N$) the weight of evidence w_j ($j=1,\dots,J$)
1: Function EMF approach (S_i, V_i)
2: Obtain the ground-truth label Y_i of the review S_i by human scoring
3: Obtain the evidence e_i^{voting} related to a review S_i by Eq. (1)
4: Extract the features X_i from both the review textual and visual contents
5: Obtain the evidence $e_i^{H-feature}$ by Eq. (2)
6: Obtain the evidence e_i^{text} by Eq. (3)
7: Obtain the evidence e_i^{image} by Eq. (4)
8: Calculate the contextual reliability of the evidence r_j ($j=1,\dots,J$) by Eq. (5)
9: Repeat until convergence
10: Initialize the weight of evidence w_j with the reliability r_j
11: For $i \in \{1,\dots,N\}$ do
12: Obtain the predicted review helpfulness values P_i ($i=1,\dots,N$) by Eq.(6)
13: Update the weight w_j ($j=1,\dots,J$)
14: End for
15: Compute the loss by Eq. (7)
16: End repeat
17: Output the predicted review helpfulness values P_i and the weight w_j ($j=1,\dots,J$)

4 Experiments

4.1 Data description

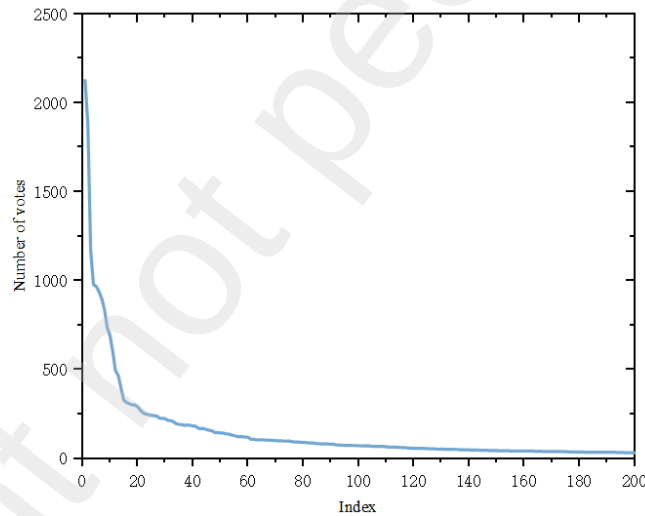
The data for experiments is collected from Amazon.com by a crawler program. Amazon reviews have been widely used as a reliable data source. Each review contains product specific meta-data including product name, reviewer name, star ratings, review titles, review date, review texts, videos, images, received votes and so on. The consumers' reviews related to a specific product category, such as smart phones, are selected for experiments. A crawler program is designed to download the information shown in the product pages from the Amazon.com website. In total, 109970 reviews were crawled in November, 2021. There are 4917 reviews with images and 558 reviews with videos. The reviews with images are selected for the experiment. 3922 reviews are obtained after deleting the review with null values and the duplicate texts. There are totally 8386 images totally. Each review has more than two images on average.

The imbalance vote bias, the winner cycle bias, and early bird bias in Amazon voting mechanism mentioned in previous literature are also identified in this experiment. There are totally 41191 votes for these reviews, concentrated on 53.3% reviews. It means that 46.7% reviews do not receive votes, as shown in **Table 3**. Among the voted reviews, 73.68% reviews received no more than 10 votes. Meanwhile, there exists great difference among helpful votes. The number of votes on each review ranged from 0 to 2126. It indicates that a larger proportion of votes has concentrated on a small proportion (14.02%) of reviews. Therefore, the imbalance vote bias is consistent with the findings in Liu et al. (2008) and Guo et al. (2021).

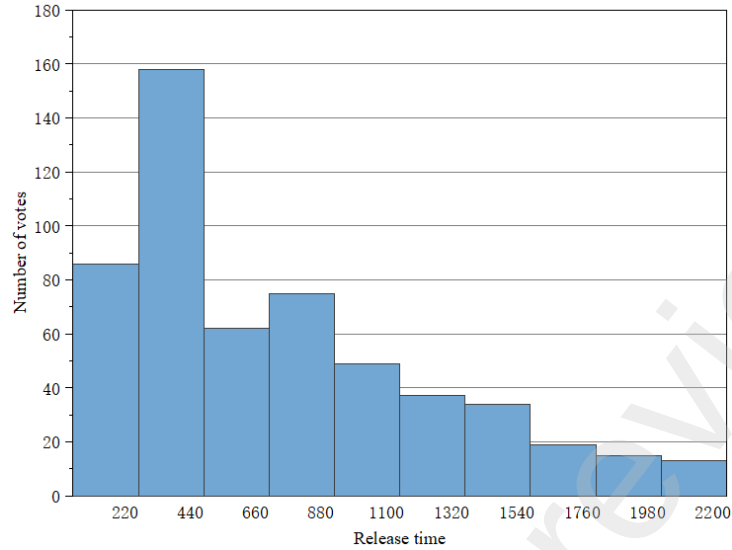
Table 3. The statistics of votes for reviews

	Review with votes		Review without votes	Total reviews
	≥ 10	< 10	0	
Review counts	550	1540	1832	3922
Percentage	14.02%	39.27%	46.71%	100%

The winner cycle bias and early bird bias are also identified, as shown in **Figure 2**. The number of helpful votes for the top 200 ranked reviews decreased rapidly. The top two reviews in the ranking hold more than 2126 and 1876 votes respectively. The number of votes decrease exponentially with the low rankings. The winner cycle bias arises since the reviews with high helpfulness votes continue to be top ranked and attract more votes than others, as shown in **Figure2(a)**. The early bird bias is described in **Figure 2(b)**. The number of votes varies greatly in different release time intervals. The time intervals represent the days elapsed after the review is posted. The reviews which elapsed days are between 220 and 440 received more votes than others. Meanwhile, the reviews which elapsed days are less than 44 days do not receive any votes.



(a) The number of helpful votes for the top 200 ranked reviews (winner cycle bias)



(b) The number of helpful votes in different release time intervals (early bird bias)
Figure 2. A winner cycle bias and an early bird bias

4.2 User study

To further evaluate the performance of the proposed EMF approach, human scoring is conducted as a ground truth for the consumers' perceived helpfulness of reviews. The helpfulness degrees in user study are measured by helpful scores and used as ground-truth labels for training three decision models and the proposed EMF approach.

Ten different annotators are invited to score each review in the range from 0 to 10, according to their answers to the question "To what extent can the review help you make a purchase decision". The scores are categorized into four intervals according to the four categories of review quality defined in Liu et al. (2007). (1) unhelpful reviews with a score interval $[0,1]$. When the score is within a value range of $[0,1]$, it indicates that the review has no valuable information related to products and is not at all helpful to support purchase decision-making. (2) a little helpful reviews with a score interval $[2,4]$. When the review presents several features related to products, the score is between 2 and 4. (3) much helpful reviews with a score interval $[5,7]$. When the review includes some product features, delivery or service information, the score is between 5 and 7. (4) greatly helpful reviews with a score interval $[8,10]$. When the review describes many product-related features and has objective comments, the score is in the range from 8 to 10.

The ground truth of review helpfulness is the average scores of each review coming from the ten annotators. A threshold $[2,10]$ is set for judging whether the review is helpful or not. The reviews with score values less than 2 are regarded as unhelpful and the others are helpful. Totally there are 2613 helpful reviews and 1309 unhelpful ones used for experiments. The samples are categorized into two groups, 70% data for training and 30% for testing.

4.3 Experimental procedures

4.3.1 Network settings

The BERT adopted in the word embedding layer provides pre-trained vectors of 768 dimensions generated by a BERT base model. The WordPiece embedding with a 30,000 token vocabulary is used for word representation. The maximum number of tokens in each review is set to be 400 for improving

computation efficiency. One token is corresponding to a word in a review. That means the early 400 words in a review is retained for training and testing. The Bi-LSTM network transforms the input word embedding into a low-dimensional vector. The dimension of hidden layer is 256. The learning rate is 0.0001, the batch size is 2, the number of layers is 1, and the dimension of output vector is 256. The pre-trained VGG16 is used to obtain the image object-level representation. Then a fine-tuning VGG16 network is developed to learn the representation of images related to reviews. The batch size is 32, and learning rate is 0.0005. The dimension of output vectors is 256.

4.3.2 Baseline approaches

Both the end-to-end deep learning algorithms and aggregating models for predicting review helpfulness are compared with the proposed approach in two experiments, respectively. The common metrics including accuracy, recall, precision, AUC, and F1-measure are adopted to evaluate the prediction performance of the approaches.

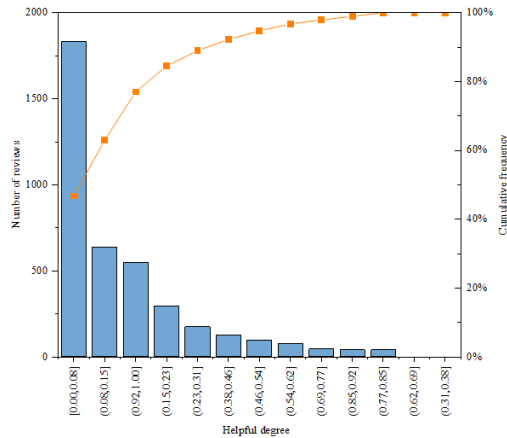
Four end-to-end deep learning algorithms involve the CNN (Olmedilla et al., 2022), Bi-LSTM, concatenation-based fusion, and attention-based fusion in the first experiment. First, both the CNN and Bi-LSTM deep learning networks are used for visual and textual feature learning and predicting review helpfulness, respectively. Then, the representation features of the two modalities are concatenated into a vector and to generate the predicted review helpfulness (Ma et al., 2018). Next, the interaction between the images and texts related to a review are considered and attention mechanism is adopted for fusing the representation features of the two modalities (Xiao et al., 2022).

Then, three aggregating models including majority voting, weighted average approach, and the ER fusion approach, are adopted for combine machine and human intelligence compared with the proposed approach in the second experiment. Majority voting approach sums the predictions from each modality and predicts the class label with the most votes (Morvant et al., 2014). Weighted average fusion approach has good rapidity and involves the importance of decision models. The fusion quality depends on the choice of weight values to a great extent (Shutova et al., 2016). In this study, the prediction accuracy of each intelligence is set to be the weight. The ER fusion approach combines multiple pieces of evidence from all decision models by the ER rule. Both the weight and reliability of evidence are set equally to be the prediction accuracy of each intelligence.

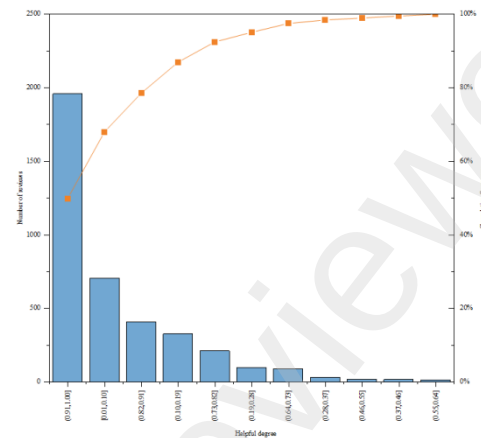
5 Results and discussion

5.1 The distribution of helpfulness degrees

The distribution of helpfulness degree measured by user votes and that predicted by the proposed EMF approach are compared to validate the effectiveness of hybrid intelligence. The helpfulness degree is firstly measured by helpful votes in voting mechanism, as shown in **Figure 3(a)**. Since there exists a large variation among the votes received by reviews, the review helpfulness degree is defined as a value with a range in $[0,1]$. The average value of the helpfulness degrees measured by initial votes is 0.25. It indicates that the values of these reviews are seriously underestimated since the 46.7% of reviews do not receive any vote. When the helpfulness degrees of reviews are estimated by the proposed EMF approach, the average value is greatly increased from 0.29 to 0.64, as shown in **Figure 3(b)**. Since the proposed approach is trained by the helpful degrees measured by votes in user study, it also means that many greatly helpful reviews neglected in voting mechanism can be identified by the proposed approach.



(a) The distribution of helpfulness degrees measured by user votes



(b) The distribution of helpfulness degrees predicted by the EMF approach

Figure 3. The distribution of helpfulness degrees in different measures

5.2 Comparison of different approaches

The helpfulness degrees of reviews are predicted by aggregating the evidence from both human and machine judgment perspectives in this study. The evidence is obtained from different decision models and combined by the ER rule for the final helpfulness degree. The reliabilities of four kinds of evidence are calculated according to the Eq. (5), as shown in **Table 4**. It indicates that the evidence from textual contents of reviews is the most reliable among them. The evidence from consumers' voting behavior has the lowest reliability since there exists three kinds of bias in voting mechanism and some greatly-helpful reviews in user study do not receive any vote in current voting mechanism. Therefore, the evidence from the initial votes is less credible for measuring the helpfulness degree of a review.

Table 4 The reliability of evidence obtained from different decision models

Judgment	Sources	Decision models	Reliability of evidence
Human judgment	Votes	Human scoring	0.5367
	Theoretical features	Logistic regression	0.8199
machine judgment	Texts	Bi-LSTM	0.8360
	Images	CNN	0.6720

The EMF approach is used for combining the evidence from different presentation formats to achieve a new helpfulness degree. It is compared with four end-to-end review helpfulness prediction approaches in a deep learning framework. The metric values of prediction performance are shown in **Table 5**. Results show that the proposed EMF approach outperforms other approaches owing to its capability of combining the decision results with uncertainties. It also indicates the necessity of aggregating both machine intelligence from the images and texts, and human judgment from user votes. Moreover, the performance of the proposed multimodal fusion approach outperforms the individual textual model Bi-LSTM and visual modal CNN.

Table 5 Prediction performance of different end-to-end deep learning approaches

Approaches	Accuracy	Precision	F1-measure	AUC
Bi-LSTM	0.8343	0.8717	0.8756	0.8774
CNN	0.6720	0.6990	0.7836	0.6134
Concatenation-based fusion	0.8292	0.8713	0.8719	0.8581
Attention-based fusion	0.8504	0.8457	0.8828	0.9213
The proposed EMF	0.8768	0.8940	0.9091	0.9216

Meanwhile, different aggregation approaches are compared with the proposed EMF approach. The metrics are shown in **Figure 4**. Obviously, the EMF approach achieved good performance. The predict

performance of the majority voting approach is very low, especially in terms of the AUC value. It means that the prediction results from different intelligence for review helpfulness have conflicts and do not agree with other prediction results.

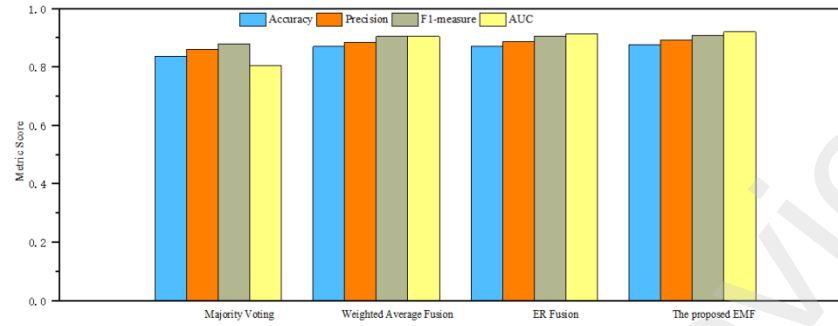


Figure 4 Prediction performance of different aggregation approaches

5.3 Complementary strengths of human and machine intelligence

The above findings show that hybrid combinations of human and machine intelligence lead to a higher performance than individual intelligence. Their relative contributions on predicting review helpfulness are visualized in Figure 5. Suppose that the judgment from both human and machine algorithms are fully reliable. The final helpfulness degree P_i can be formulated as a weighted fusion of all the decision results from each classifier. The weighted fusion process is conducted for both helpful and unhelpful reviews, respectively. For the unhelpful reviews, human judgment from votes (red blocks in Figure 5) has the most significant impact on the helpfulness of reviews. This can be also validated that there are many unhelpful reviews with zero votes indeed. For the helpful reviews, machine judgment from handcraft features, texts and images plays more important roles than human votes. This may be caused that a great number of greatly helpful reviews do not receive any user votes. It also suggests that the multimodal data sources should be adopted for calibrating the biased helpfulness degrees that measured only by user votes.



Figure 5 Complementary strengths of human and machine intelligence

5.4 Sensitivity analysis

Since the reliability of human judgment is dependent on the collected dataset in different release date, its effect on the prediction performance of the proposed EMF approach is further explored. The reliability of human judgment is manually set from zero to one with 0.1 as interval. The AUC values of the proposed EMF approach, the original ER fusion approach, and the weighted average fusion approach are summarized in Figure 6. Compared with the original ER fusion approach, the best prediction

performance of the proposed EMF approach is achieved when the reliability of human judgment equals to 0.9. Owing to the long-tailed distribution of user votes, the AUC performance will be declined to 75% even if the human judgment is fully reliable. However, the performance of weighted average fusion approach declined when the reliability of human judgment increased from 0 to 1. This may because that the decision results from user votes in this experiment are greatly deviated from the ground truth. It also validated that the human scoring is necessary for training the review helpfulness prediction models owing to the existence of vote biases.

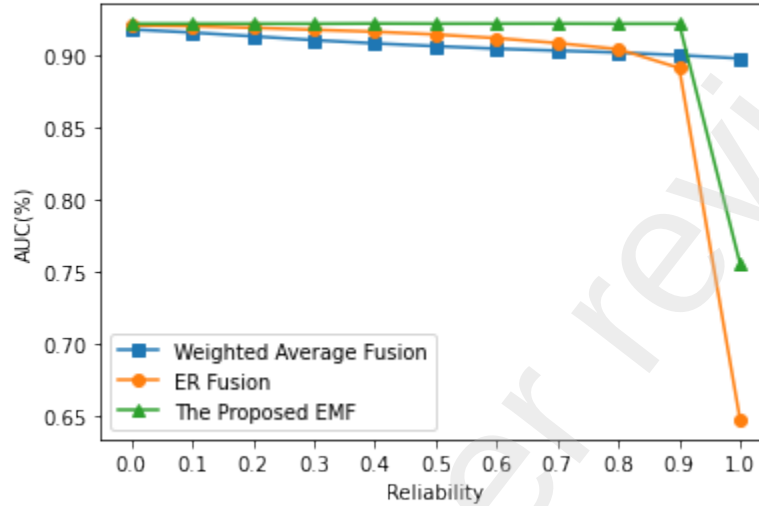


Figure 6 Prediction performance in different-level reliability of human judgment

6 Conclusion

Voting mechanism in online contexts encourages consumers to evaluate the helpfulness of reviews by helpful votes. Many techniques and models are developed and trained by the votes to automatically assess the helpfulness levels. Since there exist three kinds of bias, the prediction performance is limited when only the user votes are adopted to assess the quality of reviews. We proposed a novel evidence-based multimodal fusion approach to predict the helpfulness degrees by aggregating all the evidence related to reviews. The evidence is not only obtained from human judgment, but also from three kinds of machine judgment involving review texts, images and theoretically-motivated features. The EMF approach is developed for combining the evidence from different presentation formats for predicting the helpfulness of reviews comprehensively. A new dataset is developed for the task of multimodal review helpfulness prediction by crawling data from Amazon.com. Human scoring is conducted to label reviews and avoid the voting bias. The effectiveness and robustness of the approach are evaluated on the dataset. Experiment results show that integrating multimodal information from various presentation formats of reviews and aggregating hybrid intelligence can significantly improve the prediction accuracy.

The proposed EMF approach can enhance the helpfulness prediction performance by combining human intelligence and machine intelligence in a comprehensive way. It can improve the accuracy, precision, recall, AUC and F-measure significantly when compared with other multimodal fusion methods. When the multimodal information of reviews, including texts and images, are fused by the EMF approach to predict helpfulness degrees, the new helpfulness degrees have a higher average value than the traditional one measured only by user votes. It indicates that many greatly-helpful reviews that do not receive any votes can be identified by the proposed EMF approach.

For e-commerce platforms like Amazon.com or Jindong.com, this research contributes an enhanced method for evaluating the really-helpful reviews and providing high quality reviews for potential consumers practically. Not only the value of user-generated contents but also the content of images related to products are exploited by the proposed EMF approach. Moreover, it is obvious that the textual

contents of reviews play a prominent role in predicting the helpfulness of reviews, especially when the review has not received any vote. Therefore, both the texts and images should be encouraged by e-commerce platforms for consumers to express their opinions about products or services.

Academically, the proposed approach provides an evidence-based multimodal fusion framework for predicting helpfulness degrees of reviews. Traditional prediction techniques and models for review helpfulness mainly depend on machine intelligence supposing that the user votes are fully reliable for measure the quality of reviews. The proposed evidence-based measure can define the helpfulness degrees from various presentation formats of reviews in a general way by a belief function. Therefore, the helpfulness degrees can be measured not only by helpful votes but also by the textual and visual contents related to reviews. The proposed EMF approach can aggregate hybrid human and machine intelligence for predicting review helpfulness by exploiting user votes, texts and images related to a review. Empirical study on the Amazon.com dataset also validates the effectiveness and robustness of the approach.

Future research may focus on the extension of data sources and multimodal fusion mechanism. In this study, only the reviews related to smart phones on the Amazon.com datasets are selected as a specific category for the experiment. Other product categories and other data sources will be adopted for further comparison. Moreover, the data used for training and testing was collected at one-time point when many reviews with short release interval time cannot receive deserving votes. Since helpful votes accumulate over time, it suggests that the data would be collected in two stages of time and there is a time lag between the training and testing sets. Additionally, although the complementarity between human and machine intelligence are investigated in this study, the interaction between image and text modality data has not been studied in deep learning processes.

Acknowledgement

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